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Inflator having an outlet rotatable relative to a handle

Abstract

An inflator includes a housing, a handle formed by the housing, a motor housed in the housing, a trigger configured to activate the motor, an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor, an outlet which is transverse to the inlet and a fan configured to draw air in through the inlet and blow air out through the outlet. The inlet is generally transverse to the handle. The position of the outlet is rotatable relative to the handle.

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Application No. 63/126,133 filed on Dec. 16, 2020, entitled High Volume, Low Pressure Inflator. The entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The present disclosure relates to an inflator. In particular, exemplary embodiments disclose a high flow, low pressure inflator.

SUMMARY OF EMBODIMENTS OF THE INVENTION

(2) Aspects of the present disclosure relate to an exemplary embodiment of an inflator power tool.

(3) According to one aspect, there is an exemplary embodiment of a high-volume, low pressure inflator that includes a housing; a handle formed by the housing; a motor housed in the housing; a trigger configured to activate the motor; an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor; an outlet which is transverse to the inlet; a fan configured to draw air in through the inlet and blow air out through the outlet.

(4) The inlet may be generally transverse to the handle.

(5) The position of the outlet may be rotatable relative to the handle.

(6) The outlet may be rotatable about a rotation axis.

(7) The rotation axis may be generally concentric or parallel to the longitudinal axis of the motor.

(8) The rotation axis may be generally transverse to the handle.

(9) According to another aspect, there is an exemplary embodiment of an inflator, the inflator including a tool base unit, the tool base unit comprising a handle, a motor and a trigger; and an inflator tool head removably attached to the tool base unit.

(10) The inflator tool head may include an air inlet, an air outlet and a fan configured to draw air in through the air inlet and blow air out through the air outlet.

(11) The inlet may be transverse to the outlet.

(12) The outlet may be rotatable with respect to the tool base unit while the inflator tool head is attached to the tool base unit.

(13) The inlet may be generally transverse to the handle.

(14) The inflator may further include an attachment holder configured to hold an inflator attachment.

(15) The attachment holder may be disposed on the inflator tool head.

(16) The inflator tool head may include a connection portion and a fan housing portion.

(17) The fan may be disposed in the fan housing portion.

(18) The fan housing portion may be rotatable with respect to the connection portion.

(19) The fan housing portion may include the air outlet.

(20) The fan housing portion may include the air inlet.

(21) The fan housing portion may be configured to be held in place in a variety of rotational positions with respect to the connection portion.

(22) The fan housing portion may be configured to be held in place by friction.

(23) The fan housing portion may be configured to be held in place by a biasing assembly.

(24) The fan housing portion may be configured to be held in place at plurality of distinct rotational positions.

(25) The plurality of distinct rotational positions may include at least four distinct rotational

positions.

(26) The connection portion may house a connector which receives drive from the motor.

(27) According to another aspect, there is an exemplary embodiment of a high-volume, low pressure inflator, the inflator including a housing; a handle formed by the housing; a motor housed in the housing; a trigger configured to activate the motor; an inflator head portion; an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor; an outlet which is transverse to the inlet; a fan configured to draw air in through the inlet and blow air out through the outlet.

(28) The inlet may be generally transverse to the handle.

(29) The position of the outlet may be rotatable relative to the handle.

(30) The inflator head portion may include a connection portion and a fan housing portion.

(31) The fan may be disposed in the fan housing portion.

(32) The fan housing portion may include the outlet and is rotatable with respect to the connection portion.

(33) The outlet may be rotatable about a rotation axis.

(34) The rotation axis may be generally concentric or parallel to the longitudinal axis of the motor.

(35) The rotation axis may be generally transverse to the handle.

(36) These and other aspects of various embodiments of the present invention, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. In one embodiment of the invention, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. In addition, it should be appreciated that structural features shown or described in any one embodiment herein can be used in other embodiments as well. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

(37) All closed-ended (e.g., between A and B) and open-ended (greater than C) ranges of values disclosed herein explicitly include all ranges that fall within or nest within such ranges. For example, a disclosed range of 1-10 is understood as also disclosing, among other ranges, 2-10, 1-9, 3-9, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of embodiments of the present invention as well as other objects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

(2) FIG. 1 is a perspective view of an inflator according to a first exemplary embodiment of the present application;

(3) FIG. 2 is a perspective view of the exemplary embodiment of the inflator with the inflator tool head removed from the tool base unit;

(4) FIG. 3 is a perspective view of the inflator inflating a mattress;

(5) FIG. 4 is a perspective view of the inflator deflating the mattress;

(6) FIG. 5 is a perspective view the inflator tool head of the exemplary embodiment of the inflator;

(7) FIG. 6 is another perspective view the inflator tool head of the exemplary embodiment of the inflator;

- (8) FIG. 7 is a perspective view of an inflator according to a second exemplary embodiment of the present application;
- (9) FIG. 8 is a perspective view of the second exemplary embodiment of the inflator with the inflator tool head removed from the tool base unit;
- (10) FIG. 9 is a perspective view of the second exemplary embodiment of the inflator inflating a mattress;
- (11) FIG. 10 is a perspective view of the inflator deflating the mattress;
- (12) FIG. 11 is a perspective view the inflator tool head of the exemplary embodiment of the inflator;
- (13) FIG. 12 is another perspective view the inflator tool head of the exemplary embodiment of the inflator;
- (14) FIG. 13 is a perspective view of an inflator according to a third exemplary embodiment;
- (15) FIG. 14 is another perspective view of the inflator according to the third exemplary embodiment;
- (16) FIG. 15 is another perspective view of the inflator according to the third exemplary embodiment;
- (17) FIG. 16 is another perspective view of the inflator according to the third exemplary embodiment;
- (18) FIG. 17 is a close-up cut-away view of the inflator according to the third exemplary embodiment;
- (19) FIG. 18 is another close-up cut-away view of an inflator according to the third exemplary embodiment;
- (20) FIG. 19 is a perspective view of the inflator with attachment holders; and
- (21) FIG. 20 is a close-up perspective view of the inflator with attachment holders.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE INVENTION

- (22) FIG. 1 illustrates a first exemplary embodiment of inflator **10**. The inflator **10** of the exemplary embodiment is a high volume, low pressure inflator of that provides a high volume of airflow for inflating air mattresses, rafts and other inflatable items. As shown in FIG. 1, the inflator **10** includes a housing **11**. The housing **11** forms a handle **12** and a motor housing portion **13**. A motor is housed in the motor housing portion **13**. There may be a transmission connected to the motor. The housing **11** also has a battery receiving portion **14** configured to receive a battery pack. In this case, the battery pack may be a power tool battery pack configured to power a variety of power tools such as drills, sanders and saws. The battery pack may be of the type shown in, for example, U.S. Pat. Nos. 7,598,705; 7,661,486; or U.S. Patent Application Publication No. 2018/0331335. U.S. Pat. Nos. 7,598,705; 7,661,486; and U.S. Patent Application Publication No. 2018/0331335 are hereby incorporated by reference. The battery pack powers the inflator **10** and particularly the motor of the inflator **10**. A user-actuable trigger **15** is housed in the handle **12**. The trigger **15** can be actuated to actuate the motor.
- (23) The inflator **10** of the first exemplary embodiment includes a tool base unit **100** and an inflator tool head **200**. As is shown in FIG. 2 of the present application, the inflator tool head **200** is removably attached to the tool base unit **100**. Other tool heads such as a drill tool head, saw tool head, sander tool head and others may also be attached to the tool base unit **100** forming a multi-headed tool system. The tool base unit **100** may be of the same design as the tool base unit **100** show in U.S. Pat. No. 9,956,677, which is hereby incorporated by reference in its entirety and the tool base unit **100** of the present application may similarly accept the variety of tool heads as discussed in U.S. Pat. No. 9,956,677.
- (24) U.S. Pat. Nos. 9,421,682 and 9,776,315 are other examples of multi-headed tool systems. U.S. Pat. Nos. 9,421,682 and 9,776,315 are hereby incorporated by reference in their entireties.
- (25) The inflator head portion **200** includes a connection portion **210**. The connection portion is configured to connect to the tool base unit and transfer power from the motor of the tool base unit

100 to the tool head portion. The inflator head **200** further includes a fan housing portion **220** which houses a fan (fan **60** in FIGS. **17** and **18** is described further below). The fan draws air in through the inlet **240** and pushes air out through the air outlet **245**. In this way, the inflator **10** can be used to inflate and deflate inflatable objects.

(26) FIGS. **3** and **4** show an example of the inflator **10** inflating and deflating a mattress **500**. In FIG. **3**, the air outlet **245** is engaged with the mattress **500** and the air inlet **240** is directed towards ambient air. In this position, air is drawn in through the air inlet **240** and directed through the air outlet **245** to the mattress **500**.

(27) FIG. **4** shows the inflator **10** in a position to deflate the mattress **500**. In FIG. **4**, the air inlet **240** is engaged with the mattress **500**. The air outlet **245** is directed to ambient atmosphere. Accordingly, the inflator **10** draws air in from the mattress **500** through the air inlet **240** and blows it out of the air outlet **245**. In this way, it deflates the mattress **500**. Although the inflator **10** is shown inflating and deflating a mattress **500** in FIGS. **3** and **4**, it is understood that the inflator **10** may similarly be used to inflate or deflate a variety of other inflatable products such as rafts, intertubes, beach balls and the like.

(28) FIGS. **5** and **6** illustrate the inflator head **200** alone. FIG. **6** illustrates a portion of the connection section **210** that interfaces with the tool base unit **100**. As shown, the connection section **210** includes an intermediate input member **215** which receives a rotary drive from the motor. In particular, the intermediate input member **215** engages with an intermediate output member which is driven by the motor. In this way, rotary drive from the motor in the tool base unit **100** is transferred to the inflator tool head **200**.

(29) As shown in FIGS. **1-6**, the air inlet **240** has an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor. The air outlet **245** is transverse to the air inlet **240**. Additionally, the air inlet **240** is generally transverse to the handle **12**.

(30) FIGS. **7-12** illustrate another exemplary embodiment of an inflator **101**. The inflator **101** is the same as inflator **10** except with respect to the orientation of the air outlet **345**. In particular, the tool base unit **100** is the same as in the inflator **10**. Additionally, the inflator tool head **300** is the same as the inflator tool head **200**, except that the air outlet **345** of the inflator tool head **300** is disposed to direct air sideways/horizontal as opposed to upwards/vertical.

(31) FIG. **7** illustrates the inflator **101** in the assembled condition with the inflator tool head **300** attached to the tool base unit **100**. As shown in FIG. **7**, when the tool head **300** is attached to the tool base unit **100**, the air outlet **345** is directed to the side. That is, if the tool base unit **100** is placed on a horizontal surface, the air outlet **345** is generally to the side rather than up or down. Put another way, the air outlet **345** is generally transverse to the handle **12**. The air outlet **345** is also generally transverse to a longitudinal axis of the motor. The motor produces a rotary drive along the longitudinal axis of the motor. The longitudinal axis of the motor is generally parallel or concentric with a longitudinal axis of the motor housing **13** and is generally transverse to the handle **12**.

(32) FIG. **8** illustrates the inflator **101** with the inflator tool head **300** removed. FIG. **9** illustrates the inflator **101** with the air outlet **345** engaged with the mattress **500** in order to inflate the mattress **500**. FIG. **10** illustrates the inflator **101** with the air inlet **240** engaged with the mattress **500** in order to deflate the mattress **500**. FIGS. **11** and **12** are views of the inflator tool head **300** by itself.

(33) The first embodiment of the inflator **10** in FIGS. **1-6** illustrates the air outlet **245** in a first, upward, direction. The second embodiment of the inflator **101** in FIGS. **7-12** illustrates the air outlet **345** in a second, sideways, direction. The air outlet may also be radially located at various other directions. For example, the air outlet may be directed down instead of up. The air outlet **345** may be directed to the opposite side from that shown in FIGS. **7-12**. Additionally, the air outlets **245**, **345** may be directed at intermediate angled positions.

(34) FIGS. **13-18** illustrate another exemplary embodiment of an inflator **102**. Inflator **102** has the

same base unit **100** as the inflators **10** and **101**. Inflator **102** has an inflator tool head **400**. The inflator tool head **400** is similar to the inflator tool heads **200** and **300**. However, the inflator tool head **400** has a rotatable portion such that the air outlet **445** may be rotated 360 degrees to be pointed at various directions. For example, in FIG. **13**, the outlet **445** is pointed straight up. In FIG. **14**, the outlet **445** has been rotated to be at an angle between horizontal and vertical, roughly **45** degrees from horizontal or within a range of 30 to 60 degrees from vertical or horizontal. FIG. **15** illustrates the outlet **445** pointing to the side/horizontal and FIG. **16** illustrates the outlet pointing horizontally to the opposite side as that shown in FIG. **16**. The outlet **445** may be positioned at any angle along about the rotational axis.

(35) FIGS. **17** and **18** illustrate the rotatable structure of the inflator tool head **400**. As shown in FIGS. **17** and **18**, the inflator tool head **400** has a fan housing portion **420** that is made as a separate part and is rotatable with respect to the connection portion **410**. This is in contrast to the inflator tool heads **200** and **300**, in which the connection portion **210** and the fan housing portion **220** are made of a single piece.

(36) As shown in FIGS. **17** and **18**, the connection portion **410** has a projection **411** and the fan housing portion **420** has a groove **421**. The projection **411** engages with the groove **421** to rotatably connect the fan housing portion **420** and the connection portion **410**. As also shown in FIGS. **17** and **18**, the fan connection portion **410** has an abutting wall **412** and the fan housing portion has an abutting wall **422**. The connection portion abutting wall **412** abuts the housing portion abutting wall **422** providing stability to the connection. In some embodiments, these abutting walls **412**, **422** provide friction to maintain the relative rotational location of the fan housing portion **420** to the fan connection portion **410**. The user can then rotate the fan housing portion **420** by applying pressure to overcome the friction. In this way, the user can change the angular position of the air outlet **445** among the various rotational locations. The amount of friction can be provided by the dimensioning of the connection and fan housing portions **410** and **420**; the materials of the connection and fan housing portions **410** and **420** or the abutting walls **412**, **422**; surface treatments of the connection and fan housing portions **410** and **420** or the abutting walls **412**, **422**; other methods for increasing or controlling friction or a combination of the same. For example, the abutting walls **412**, **422** may be made to tightly abut one another in order to increase friction. As another example, facing surfaces of the abutting walls **412**, **422** may be coated with a high friction material such as rubber in order to increase the friction between the connection housing **410** and fan housing portions **420**.

(37) The fan housing portion includes both the air inlet **240** and the air outlet **445**. The air inlet **240** is at or on a rotational axis of the fan housing portion **420**. Accordingly, although the air inlet **240** of the exemplary embodiment rotates along with the fan housing portion **420**, the axis of the air inlet **240** changes none if on the rotational axis or only slightly if slightly off of the rotational axis. Thus, the overall position of the air inlet **240** is relatively the same in all positions of the fan housing portion **420**. In other exemplary embodiments, the air inlet **240** may be disposed offset from the rotational axis of the fan housing portion **420** such that the position of the axis of the air inlet **240** also changes when the fan housing portion **420** rotates.

(38) In contrast, the air outlet **445** is at an outer circumference of the fan housing portion **420**. Additionally, the air outlet **445** has an axis that is transverse to a rotational axis of the fan housing portion **240**. Accordingly, the position of the air outlet **445** changes along with the rotational position of the fan housing portion **240**.

(39) The rotational position of the fan housing portion **420** may alternatively or additionally be provided by a biasing assembly. In the exemplary embodiment, there is ball **430** biased by a spring **431** from the connection portion **410** towards the fan housing portion **420**. The ball **430** contacts the fan housing portion **420** to maintain the fan housing portion **420** in a rotational location. A user may provide a force to overcome the force provided by the ball **430** to move from the particular rotational location to another rotational location. The fan housing portion **420** may include recesses

at particular locations that receive the ball **430**. The recesses **435** are located at particular specific locations which may be useful to a user. For example, the recesses **435** may be placed at four locations about the fan housing portion **420**, ninety-degrees apart. That would allow the fan housing portion **420** to be specifically set at those four specific and discrete locations, such as up, down, left and right. In other embodiments, there may be eight recesses **435**, each forty-five degrees apart. In that instance, four additional locations could be provided between the up, down, left and right positions mentioned above. There may be more or less recesses **435** providing a fewer or greater number of discrete locations for the disposition of the fan housing portion **420** and thus the air outlet **445**. There may also be more than one biasing assembly. Additionally, the biasing assembly may be different than a ball biased by a spring. For example, another biasing member may be used instead of a spring and instead of a ball there may be another type of projection or rigid element that engages with the recesses **435**. The biasing member and rigid element may also be one piece so that the biasing assembly is a one piece member that serves the same function. For example, this could be a compressible projection.

(40) Accordingly, as shown in FIGS. **12-18**, the position of the air outlet **445** is rotatable relative to the handle **12** and is rotatable about a rotation axis. As shown, the rotation axis may be generally concentric or parallel to the longitudinal axis of the motor and generally transverse to the handle **12**.

(41) FIGS. **17** and **18** provide cross-sectional views of the inflator head **400** so that internals of the inflator head **400** can be seen. As shown in FIGS. **17** and **18**, the inflator head **400** includes a drive connector **70**. The drive connector **70** receives rotational motion from the motor. A spindle **75** is driven through the drive connector **70** and is connected to a fan **60**. Accordingly, the motor drives the fan **60**. Internals of the inflator heads **200** and **300** are the same as that for the inflator head **400** except that the inflator heads **200** and **300** do not include the rotational features of inflator head **400**. Accordingly, inflator heads **200** and **300** similarly include drive connector **70**, spindle **75** and fan **60**. Groove **421**, projection **411** and abutting walls **412**, **422** and the biasing assembly may be removed in inflator heads **200** and **300**. However, one or more of these components may also be maintained in the inflator heads **200**, **300** in order to provide rigidity. For example, the structure of abutting walls **412**, **422** may be included in the inflator heads **200**, **300** in order to provide rigidity for the inflator heads even if they lack a rotation feature. In that case, the abutting walls **412**, **422** may be joined or formed as one integral part.

(42) FIGS. **19** and **20** illustrate inflator attachment holders **500**. The inflator attachment holders **500** are attached to either side of the connection portion **410**. The inflator attachment holders **500** hold inflator attachments **510** which can be slid over the air inlet or air outlet so as to interface with various inflatables. FIG. **19** illustrates the attachment holders **500** holding the inflator attachments **510** and FIG. **20** illustrates the attachment holders **500** when empty.

(43) In the exemplary embodiments, the inflators **10**, **101** and **102** are formed by a tool base unit and a removable inflator tool head unit. The inflators may also be made as a stand-alone tool where the inflator tool head is integrated with a base unit in a fixed manner rather than being separable.

(44) Although the present technology has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred embodiments, it is to be understood that such detail is solely for that purpose and that the technology is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present technology contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

(45) Additionally, while the exemplary embodiment is described with respect to an inflating tool, the methods and configurations may also apply to or encompass other power tools.

Claims

1. An inflator, comprising: a tool base unit, the tool base unit comprising a handle, a motor and a trigger; and an inflator tool head removably attached to the tool base unit; wherein the inflator tool head comprises an air inlet, an air outlet and a fan configured to draw air in through the air inlet and blow air out through the air outlet; wherein the inflator tool head comprises a connection portion and a fan housing portion; wherein the fan is disposed in the fan housing portion; and wherein the fan housing portion is rotatable with respect to the connection portion.
 2. The inflator according to claim 1, wherein the fan housing portion includes the air outlet.
 3. The inflator according to claim 2, wherein the fan housing portion includes the air inlet.
 4. The inflator according to claim 2, wherein the fan housing portion is configured to be held in place in a variety of rotational positions with respect to the connection portion.
 5. The inflator according to claim 4, wherein the fan housing portion is configured to be held in place by friction.
 6. The inflator according to claim 4, wherein the fan housing portion is configured to be held in place by a biasing assembly.
 7. The inflator according to claim 6, wherein the fan housing portion is configured to be held in place at plurality of distinct rotational positions.
 8. The inflator according to claim 7, wherein the plurality of distinct rotational positions comprises at least four distinct rotational positions.
 9. The inflator according to claim 8, wherein the connection portion houses a connector which receives drive from the motor.
 10. A high-volume, low pressure inflator, comprising: a housing; a handle formed by the housing; a motor housed in the housing; a trigger configured to activate the motor; an inflator head portion; an inlet having an inlet axis that is generally parallel or concentric with a longitudinal axis of the motor; an outlet which is transverse to the inlet; a fan configured to draw air in through the inlet and blow air out through the outlet; wherein the inlet is generally transverse to the handle; wherein the position of the outlet is rotatable relative to the handle; wherein the inflator head portion comprises a connection portion and a fan housing portion; wherein the fan is disposed in the fan housing portion; and wherein the fan housing portion includes the outlet and is rotatable with respect to the connection portion.
 11. The high-volume, low pressure inflator of claim 10, wherein the outlet is rotatable about a rotation axis; wherein the rotation axis is generally concentric or parallel to the longitudinal axis of the motor; and wherein the rotation axis is generally transverse to the handle.
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